Exp. No: 4	Measurement and Analysis of Dispersion Compensation in Optical Fiber using Opti System

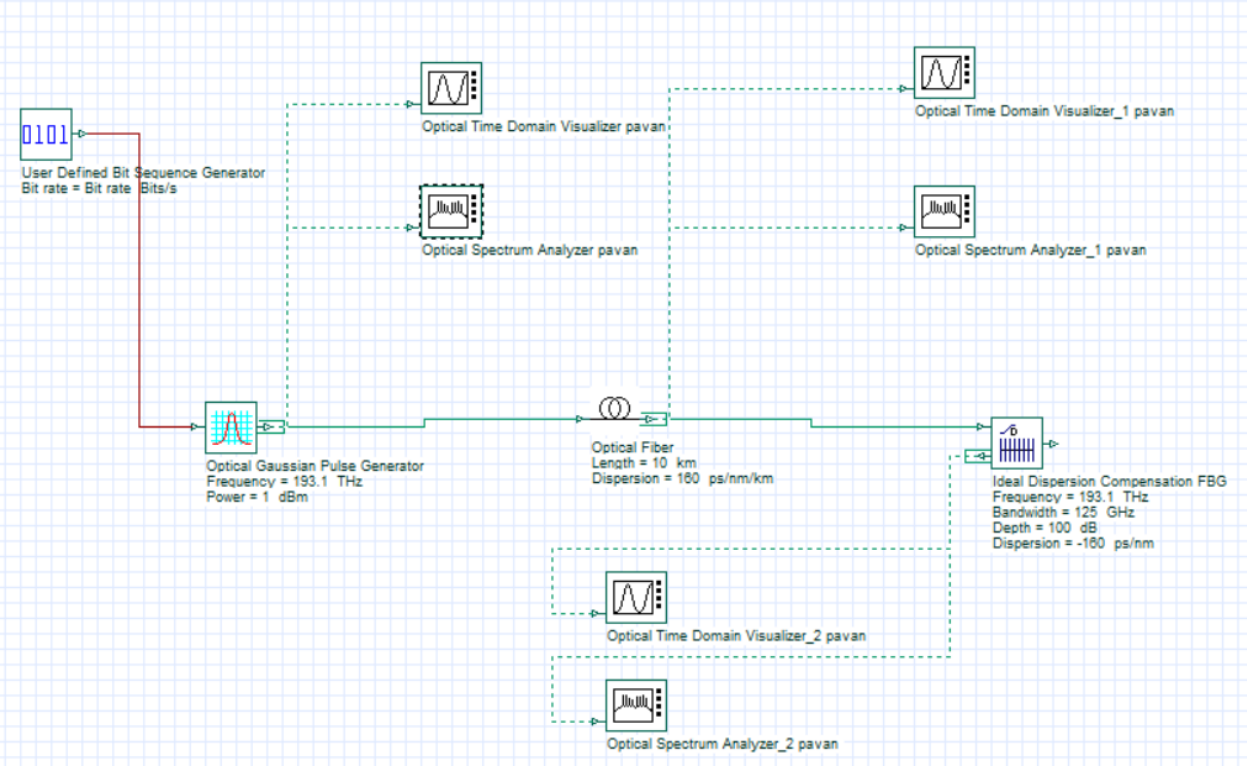
OBJECTIVES

1. To demonstrate dispersion compensation using the Ideal Dispersion Compensation component in OptiSystem.
2. To observe the broadening of a 12.5 ps optical Gaussian pulse after propagation through 10 km of single-mode fiber (SMF).
3. To compensate the accumulated dispersion of 160 ps/nm and verify recovery of the original pulse shape.

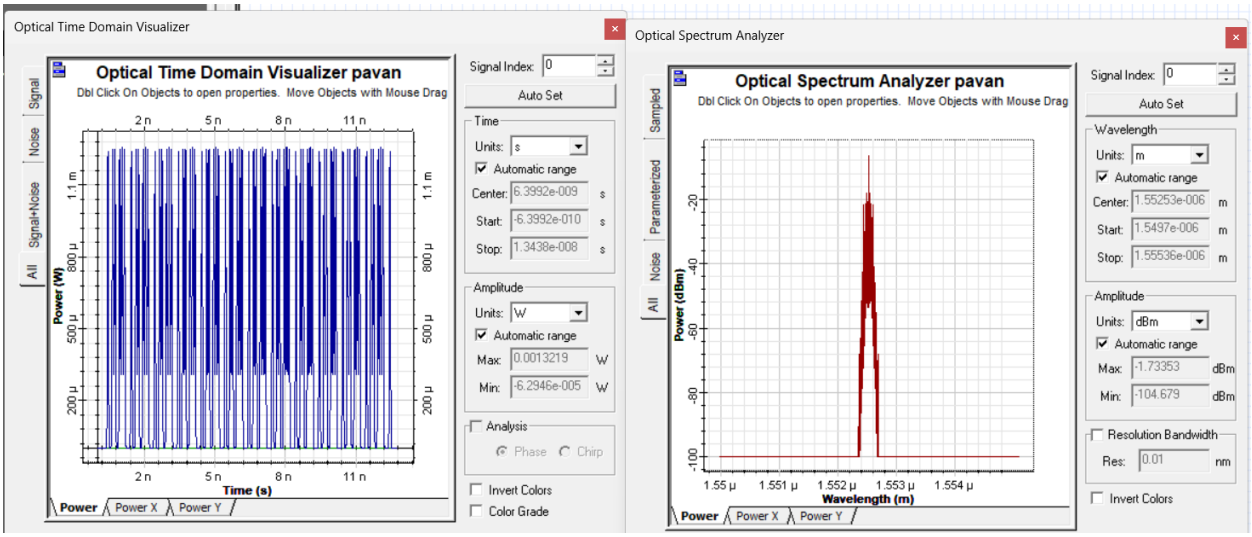
SOFTWARE / HARDWARE REQUIRED

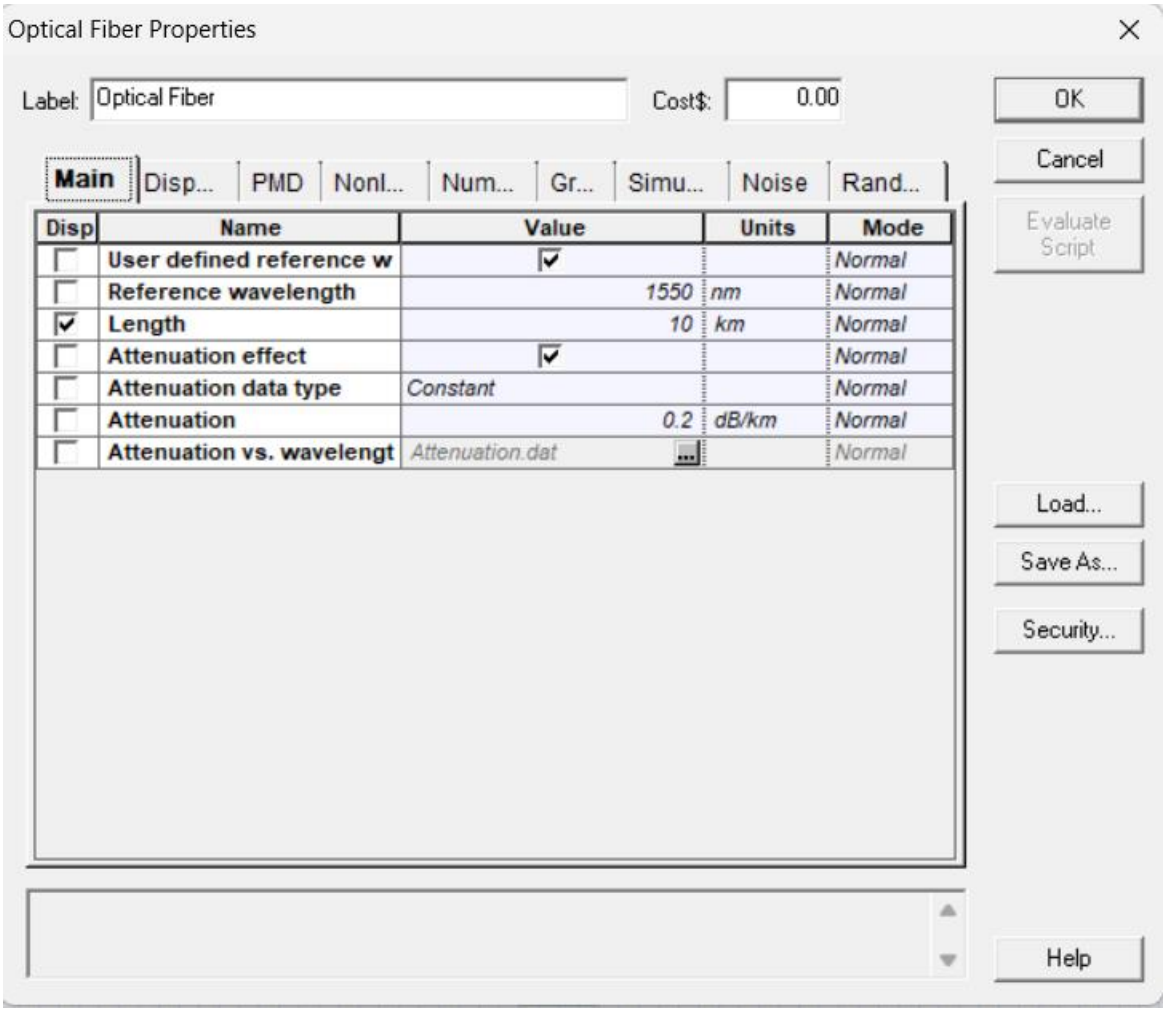
- OptiSystem Software

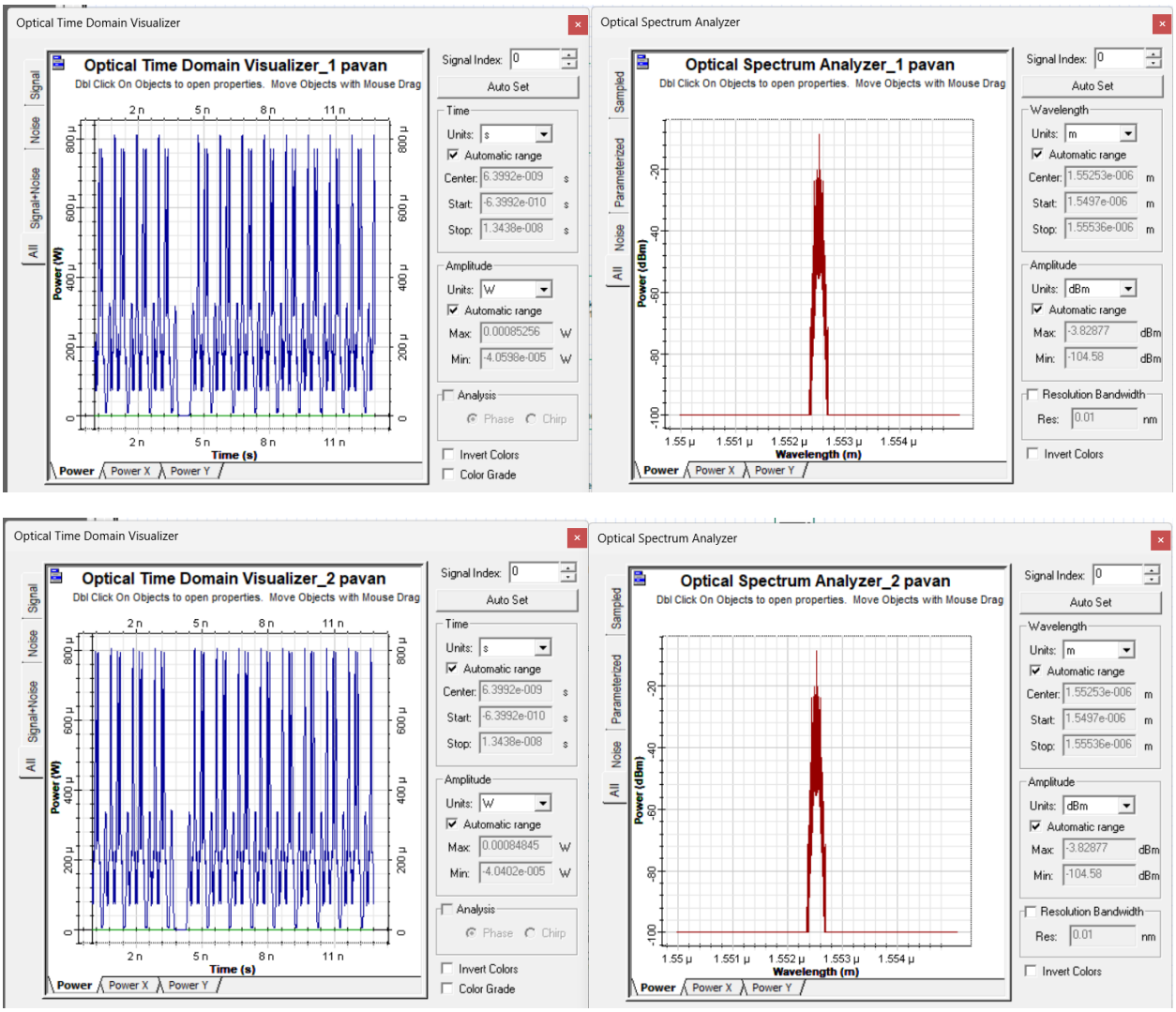
DISPERSION COMPENSATION BLOCK DIAGRAM



OUTPUTS:


Optical Time Domain Visualizer
Optical Time Domain Visualizer pavan
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Signal
Noise
Signal+Noise
All
Power (W)
Time (s)
Power
Power X
Power Y
Signal Index: 0
Auto Set
Time
Units: s
Automatic range
Center: 6.3992e-009 s
Start: -6.3992e-010 s
Stop: 1.3438e-008 s
Amplitude
Units: W
Automatic range
Max: 0.0013219 W
Min: -6.2946e-005 W
Analysis
Phase
Chirp
Invert Colors
Color Grade
Optical Spectrum Analyzer
Optical Spectrum Analyzer pavan
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Sampled
Parameterized
Noise
All
Power (dBm)
Wavelength (m)
Power
Power X
Power Y
Signal Index: 0
Auto Set
Wavelength
Units: m
Automatic range
Center: 1.55253e-006 m
Start: 1.5497e-006 m
Stop: 1.55536e-006 m
Amplitude
Units: dBm
Automatic range
Max: -1.73353 dBm
Min: -104.679 dBm
Resolution Bandwidth
Res: 0.01 nm
Invert Colors


Optical Fiber Properties
Label: Optical Fiber
Cost$: 0.00
OK
Cancel
Evaluate Script
Main
Disp...
PMD
Nonl...
Num...
Gr...
Simu...
Noise
Rand...
Disp
Name
Value
Units
Mode
User defined reference w
Normal
Reference wavelength
1550
nm
Normal
Length
10
km
Normal
Attenuation effect
Normal
Attenuation data type
Constant
Normal
Attenuation
0.2
dB/km
Normal
Attenuation vs. wavelengt
Attenuation.dat
Normal
Load...
Save As...
Security...
Help



Reflection:

1. Chromatic dispersion caused significant pulse broadening after transmission through the optical fiber.
2. The dispersion compensator effectively restored the pulse to its original shape by canceling accumulated dispersion.This experiment demonstrated the importance of dispersion compensation for maintaining signal quality in optical communication systems.

CONCLUSION:

Dispersion compensation was successfully demonstrated using the Ideal Dispersion Compensation component in OptiSystem. A 12.5 ps Gaussian pulse broadened approximately four times after 10 km of SMF (accumulated dispersion 160 ps/nm), and se ing the compensator dispersion to –160 ps/nm restored the pulse, achieving an exact compensation of the accumulated dispersion.

LABORATORY PERFORMANCE ASSESSMENT RUBRIC

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Design the OptiSystem layout	Project layout	Builds the complete layout with correct components and connections independently	Minor connection errors corrected	Needs guidance for components or connections	Unable to build the layout	20
Configure pulse, SMF and compensator parameters	Parameter configuration	Sets all parameters (40 Gb/s, 12.5 ps, 193.1 THz, 10 km, −160 ps/nm, 125 GHz) correctly	Minor parameter errors	Needs guidance for several parameters	Unable to configure parameters	20
Observe pulse broadening in SMF	Pulse observation	Captures and interprets initial and broadened pulses correctly	Minor interpretation errors	Limited observation / needs help	Unable to obtain or interpret pulses	20
Verify compensation using ideal dispersion component	Compensation verification	Verifies recovery of the pulse and explains residual effects of se ings	Mostly correct verification	Partial verification with assistance	Unable to verify compensation	20
Document findings	Report and conclusion	Complete report with justified conclusion	Minor omissions	Incomplete record / weak conclusion	Report / conclusion inadequate	20